# Reframed Systemic Problem Statement

IIIT Hyderabad Breakfast Mess System

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## The Problem, Reframed

> The Breakfast Paradox is a stable equilibrium. The system holds the gap between registration and attendance in place because every actor's behaviour inside it is individually reasonable, and every consequence it produces is absorbed before it reaches anyone who sets the rules.
>
> Three structures hold it. Breakfast is booked by default: every student is registered automatically, on campus or not, holidays included, and can leave the booking only through five cancellations per meal type per month, each made at least four days ahead. A default the system assigns cannot follow the decision each student makes on the morning. Billing fires at the booking, not at consumption, and beyond the five cancellations nothing removes the charge, so the inaccuracy costs the system nothing and, if operators are paid on booked plates, costs the operator nothing either. And the gap is measured twice, in the CDS scan record and in the operators' own books, but neither measurement is routed to the people who set registration, billing and menu rules, whose decision rights are split so that none of them holds both the knowledge and the authority to act. Lunch runs under the same booking and billing rules and mostly works, which is what makes this a breakfast problem and not a billing problem.
>
> What follows is quietly absorbed. Resale takes the student's loss. The operator's own discount on the booking, live top-ups, leftover reuse, the outside diners and staff who eat from the same food, and late-door service take most of the physical surplus. Holding and reheating turns some of the rest into lower quality, and what cannot be used leaves in the bin. Outside venues take the dissatisfaction, and informal workarounds take the cultural mismatch. Most of this is useful and none of the useful parts should be removed. Together they keep the gap from producing a signal loud enough to reach a system that responds to complaint intensity.
>
> No single actor feels this as their whole problem. The student feels a charge. One operator measures the gap every morning and has nobody to tell, another cooks to the count as the obligation it is paid for, and a third adjusts by weekly judgement but never lets its first batch fall below a floor held against shortage. The office holds a statistic. The paradox is seen in aggregate only by the actors with the least authority over the rules.

The original question was operational: students register, many do not show up, they are charged anyway and the food is cooked regardless, so why does registration not match who eats? That question sits at the wrong layer.

Beneath the kitchen half of the problem is one mechanism that is downstream of the three structures and inside authority the operator already holds. Each operator converts the booking into a cooked first batch above turnout, and can then top it up but never shrink it. Kadamba does this by a fixed ratio, Bakul by a weekly judgement with a floor, and Yuktāhār from its own books.

The rest of this document makes the case: the assumptions the statement overturns, what each stakeholder sees, the tensions that hold the system, and the evidence that validates it. The Design Opportunity Statement is a separate document that turns the diagnosis into where to intervene. Bakul has no records of any kind, so every Bakul figure is the operator's own statement.

## 1. Challenging Initial Assumptions: What the Original Question Got Wrong

The original question rested on five assumptions. Six student interviews, a CFS Chair interview, three operator interviews, a team member's first-hand account of two kitchens, Yuktāhār's books and 11,941 timestamped Kadamba scans test each one.

**Students not showing up is the problem.** It is a symptom, and part of what the turnout number counts as failure is not a problem. One respondent fasts intermittently by choice and does not notice the mess even in the same hostel. Others come from households where a formal breakfast was never the norm. The records show the same variety: at Kadamba in April, 48.5% of non-vegetarian bookings were eaten against 32.4% of vegetarian ones, and Sunday was the lowest day at Kadamba and among the lowest at Yuktāhār (confirmed). Bakul's manager calls weekends very low (single-sourced, no records). The billing record shows a considered choice and a forgotten booking identically, as a paid slot nobody used.

**Students are careless or irrational.** Most are not choosing a booking at all. As one put it: ”Even if you want to eat breakfast or not, it is automatically registered. We don't have any opportunity to nullify it.” The exit is narrow and early, so it cannot follow a morning decision (confirmed, from the CDS poster and the student's account). Kadamba bookings sat near 1,070 a day on every day of the week in April, Sunday included, while turnout ran from 42.1% on Tuesdays to 27.3% on Sundays (confirmed). Reselling on Mess Cell makes sense when Skip Meal pays nothing. Deciding at 9:10 whether the walk is worth it makes sense when the mess is about to reach its most crowded and popular items may already be gone. Every individual decision is defensible, which is why this is a systems problem.

**Nobody has solved this.** The institution has run attendance-tracked billing twice, during a gas shortage and during holidays such as Felicity. Students scanned in and were charged only if they ate. It worked both times and was switched off both times once the trigger passed, with no sign anyone evaluated keeping any of it. The data that model needs is still collected every day, one scan per breakfast eaten (confirmed), with a handwritten fallback when a scanner fails (single-sourced).

**Nobody knows how big the gap is.** It is measured in at least three places and never reconciled. Of 32,162 Kadamba breakfast bookings in April, 11,941 were eaten, 37.1%, leaving 62.9% unused. Yuktāhār's September register shows breakfast at 17.7% to 40.2%, about 28% on average, against lunch at 56% to 89% on the same days (confirmed). The CFS Chair's recalled 35% to 40% for breakfast holds for Kadamba, and his 70% for general items matches lunch. The project's earlier gap of about 30% is a lunch figure. Beliefs sit above the records: Yuktāhār's site head puts breakfast at up to about 50%, Bakul's operator believes Kadamba fills 90% to 95%, and Bakul's manager estimates at most about 60% for his own hall, an upper bound if Bakul resembles the two recorded halls (candidate). The measurement is also uneven. Yuktāhār weighs every item, Kadamba keeps gram standards and a waste estimate of 5% to 10% a day on an unstated base (single-sourced), and Bakul's lives in a weekly review that may be in one person's head. None of it has reached a registration or billing decision. The gap is known and has not travelled upward.

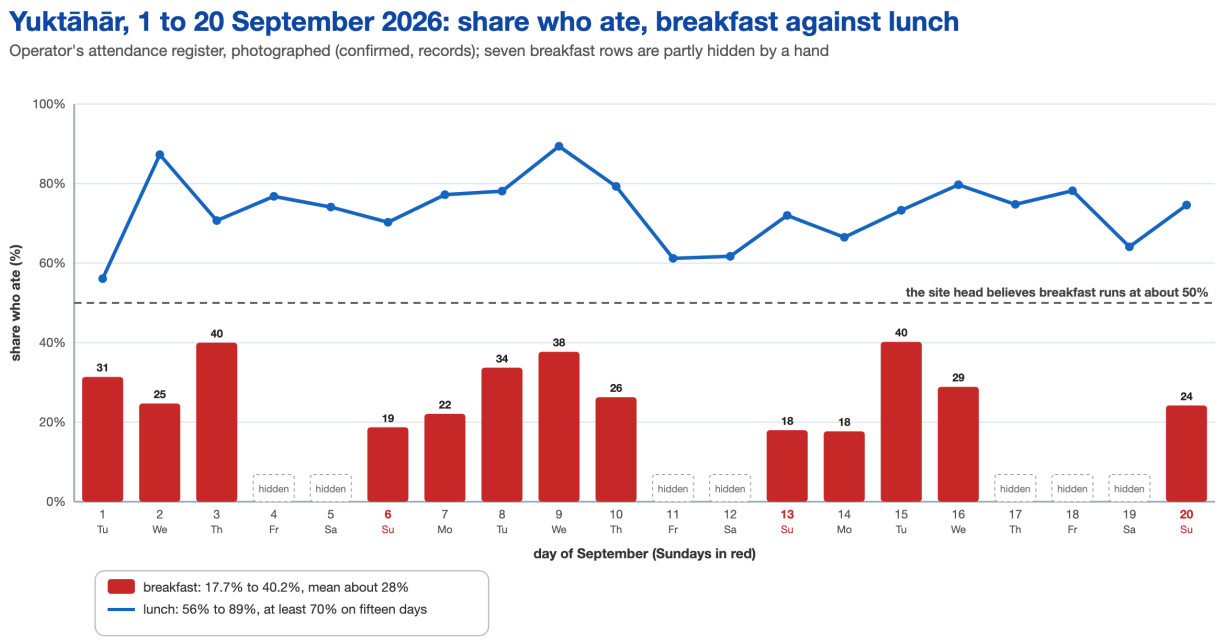



*Figure 1. Yuktāhār, September 2026: breakfast against lunch under the same booking and billing rules.*

**This is one problem.** It is three structural causes plus one thing that is not a problem at all, and treating it as one gap produces a single fix aimed at the average of four different things.

## 2. Comparing Stakeholder Perspectives: What Each Stakeholder Actually Experiences

Ask anyone in the system what their problem is and none of them will describe the Breakfast Paradox. Each feels a smaller piece, and the figure sets that beside what each cannot see.

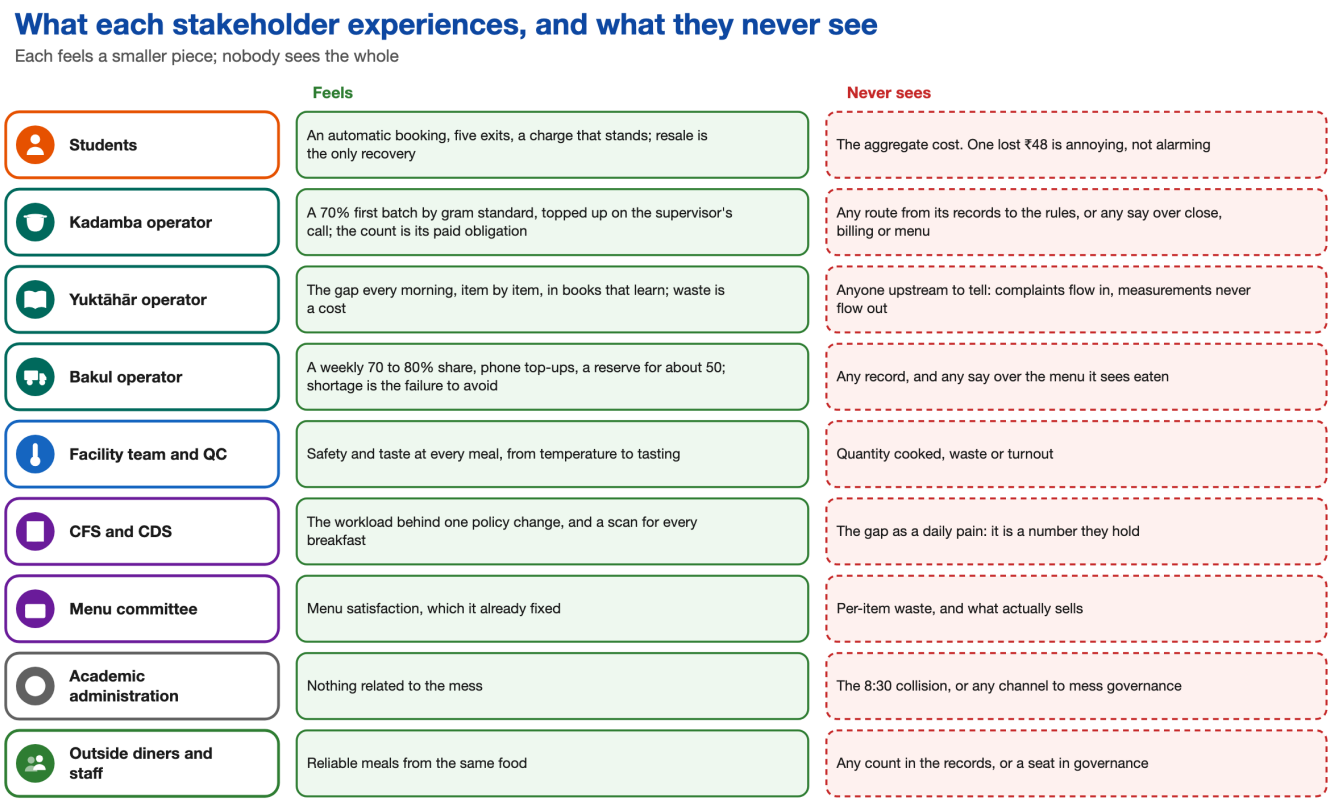



***Figure 2.*** *What each stakeholder experiences, and what stays invisible to them.*

The operators also carry their own reading of the gap. Kadamba's manager says some no-shows are students who book while away in their village while a friend eats in their place. The scan record carries no student ID and cannot test this, but automatic registration of students who are off campus would produce the pattern without anyone gaming it (candidate). Bakul's manager gives two reasons, that students may not like the menu or may not wake up (single-sourced). Not waking matches what the six interviewed students say most often. Taste is not a leading reason in those interviews, and none of the six is known to live at Bakul.

This is the centre of the reframe. The operator sees the gap more completely than anyone, and at Yuktāhār, where it is recorded per item and weekday, holds no authority over any rule that produces it. At Kadamba the operator sees live counter flow and gram standards and keeps no outcome record the team has seen. At Bakul it reads the gap through a goal that puts avoiding a shortage ahead of avoiding waste. The paradox is visible in aggregate only at the bottom of the chain of authority.

## 3. Identifying Contradictions and Tensions: What Holds This in Place

None of these tensions is fixed by asking one side to behave better. Each side is already behaving sensibly given the structure around it.

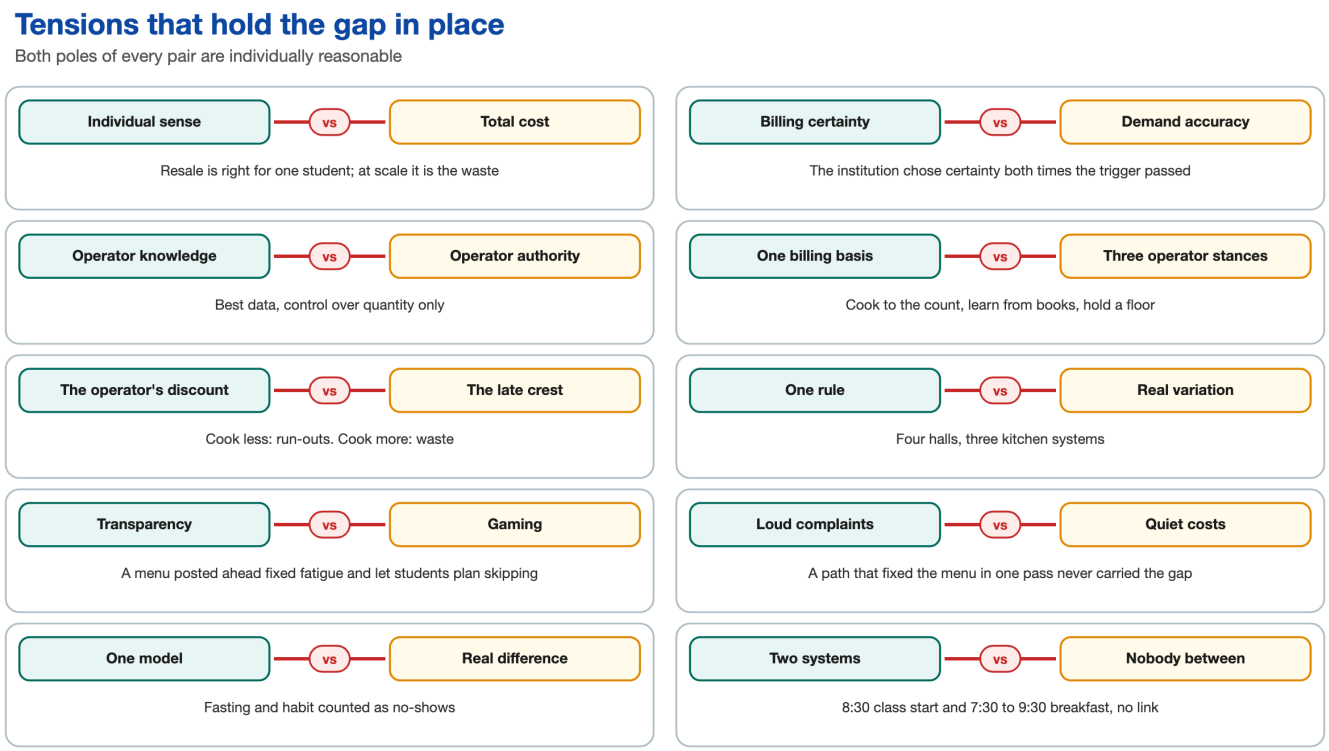



***Figure 3.*** *The tensions: each pair is rational on both sides.*

**Individual sense against total cost.** Reselling what goes unused is the right move for one student. When everyone does it, the default produces the waste and crowding this project set out to explain.

**Billing certainty against demand accuracy.** The institution has run both sides. The standing model gives it a certain revenue base fixed four days out. The attendance-tracked model matches real attendance and gives up that certainty, and the institution chose certainty each time the trigger passed. The four-day lead does not protect the kitchen. Yuktāhār buys groceries fortnightly, vegetables two or three times a week and milk daily, and Kadamba buys dry goods a week ahead, some vegetables two or three days ahead and perishables the day before. Only dry goods are committed before the close, raw surplus can be carried forward, and cooked food lasts three to four hours. The binding constraint is the cooked first batch, which the pre-service safety sequence fixes before any turnout signal arrives (candidate).

**Operator knowledge against operator authority.** The actor with the most detailed demand data controls quantity only. Yuktāhār plans each breakfast in a daily meeting from portal bookings, the Wastage Book entry for the matching week of the repeating menu, a weekday register, the month's pattern and any events, and re-projects from the first half hour of service. Its store book must match what was cooked, it holds an impurity norm for every raw item, and it dropped commercial prediction software because the data already sits in its books and its staff are not digitally inclined (single-sourced). It cannot touch the registration close, the billing basis or the menu, and exams and events reach it only if someone mentions them in the meeting. Bakul's manager wants a say in the menu and objects to pairings, not dishes: kachori belongs with chole, and he would serve Raj Kachori with aloo curry. That is his view, untested with Bakul residents, and nobody asks them.

**One billing basis against three operator stances.** All three work under the same booking and a tendered plate payment whose basis is unconfirmed (section 7). Kadamba's operator cooks to the count because ”it is a responsibility and they are getting paid for it.” Yuktāhār names lower cost and higher profit as goals and built a book system that learns. Bakul's manager states that there should not be a shortage of food even if some wastage occurs, and keeps food for about 70% of registrations on mornings he calls very low (all single-sourced). The same rule yields one kitchen that measures and trims, one that cooks a fixed ratio, and one that adjusts but only down to a floor. Whether an operator produces the information that could correct the rules depends on its culture, not on the rules

(candidate).

**The operator's discount against the late crest.** This is the one place where shortage and waste share a morning. Kadamba's first batch of 70% of registrations is set against 37.1% turnout. On April's average day that is about 530 vegetarian portions against about 245 eaten, roughly 2.2 times, and about 220 non-vegetarian against about 153, roughly 1.4 times (candidate arithmetic), so a fixed ratio over-cooks most where turnout is lowest. The operator's 5% to 10% waste figure and a first batch near twice turnout can describe the same breakfast only if outside diners, reheating or small top-ups absorb most of the difference, or the figure uses another base. That is left open. Yet late students still meet run-outs, because the crowd crests at close: 26.5% of April scans fall between 9:15 and 9:30, about nine a minute by 9:29, and 9.2%, about 37 a day, come at or after 9:30 through a back door the mess cannot refuse (confirmed; back door single-sourced). Cooking less raises run-outs and cooking more raises waste. The crest is not universal: Bakul's manager puts his rush at 8:00 to 8:30 with 10 to 15 people in the last ten minutes, and holds food for about 50 at 9:20 because a top-up phoned to a kitchen 20 to 40 minutes away cannot land in time (single-sourced; the reason is a candidate).

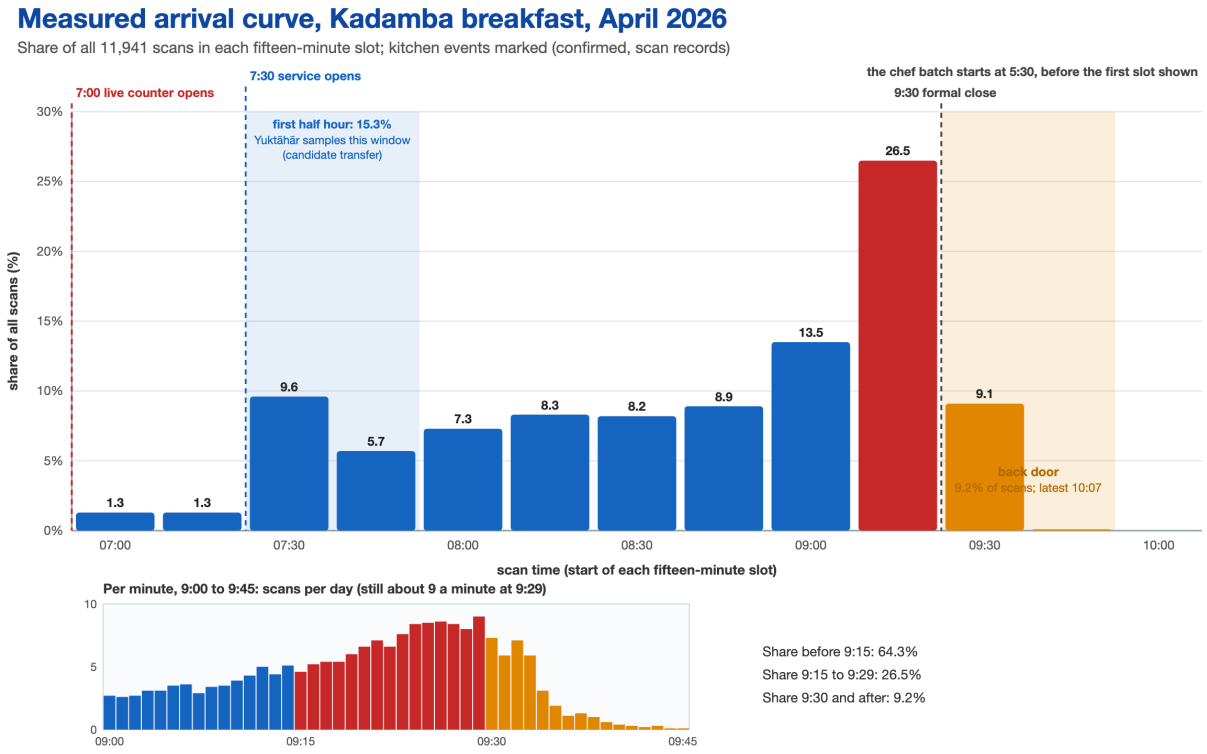



*Figure 4. Kadamba's measured arrival curve: the crowd crests at the close, and about 9% arrive after it.*

**One rule against real variation.** One registration rule and one billing rule cover four halls that run on three kitchen systems. Kadamba cooks 70% first and tops up in 10 to 15 minutes. Yuktāhār's first-half-hour projection samples the quietest part of service and so under-reads the crest (candidate). Bakul's top-up is a phone call, cooking and a drive of about 8 km, with a total lead time nobody has measured. Kadamba cooks biryani lunch at 100% of registrations, so its ratio follows the dish and not any record of turnout (candidate). Turnout varies by diet and weekday within one hall while the booking does not.

**Transparency against gaming.** Posting the menu two weeks ahead fixed menu fatigue, and by the same office's account it also let students plan their skipping. The same repeating cycle is what lets Yuktāhār look up the matching week.

**Loud complaints against quiet costs.** The path that fixed menu fatigue in one pass has never carried the larger, already measured registration gap. Complaints reach operators fast, from a director's email to the kitchen WhatsApp group, and nothing runs the other way. The facility team inspects safety and taste daily and never asks about quantity

(candidate). The same logic works inside the kitchen. At Bakul a run-out routes a complaint up to the managing director while a surplus is silent, so the operator's rule of no shortage at any cost of waste is the kitchen's form of a system that reacts only to what is loud (candidate).

**One model against real difference, and two systems with nobody between them.** Fasting, regional habits and religious observance fold into the same turnout number as ordinary no-shows. The 8:30 class start and the 7:30 to 9:30 breakfast window have no confirmed relationship, formal or informal, between their owners. The Kadamba record shows no surge before 8:30, and arrivals run at about 6% to 10% a quarter hour until 9:00. A first class ending near 9:30 could push students into the minutes the kitchen can least serve, which stays a candidate until checked against the timetable.

## 4. Reading It Through the Iceberg

The original framing stayed in the top two layers. The causes sit in the bottom two.

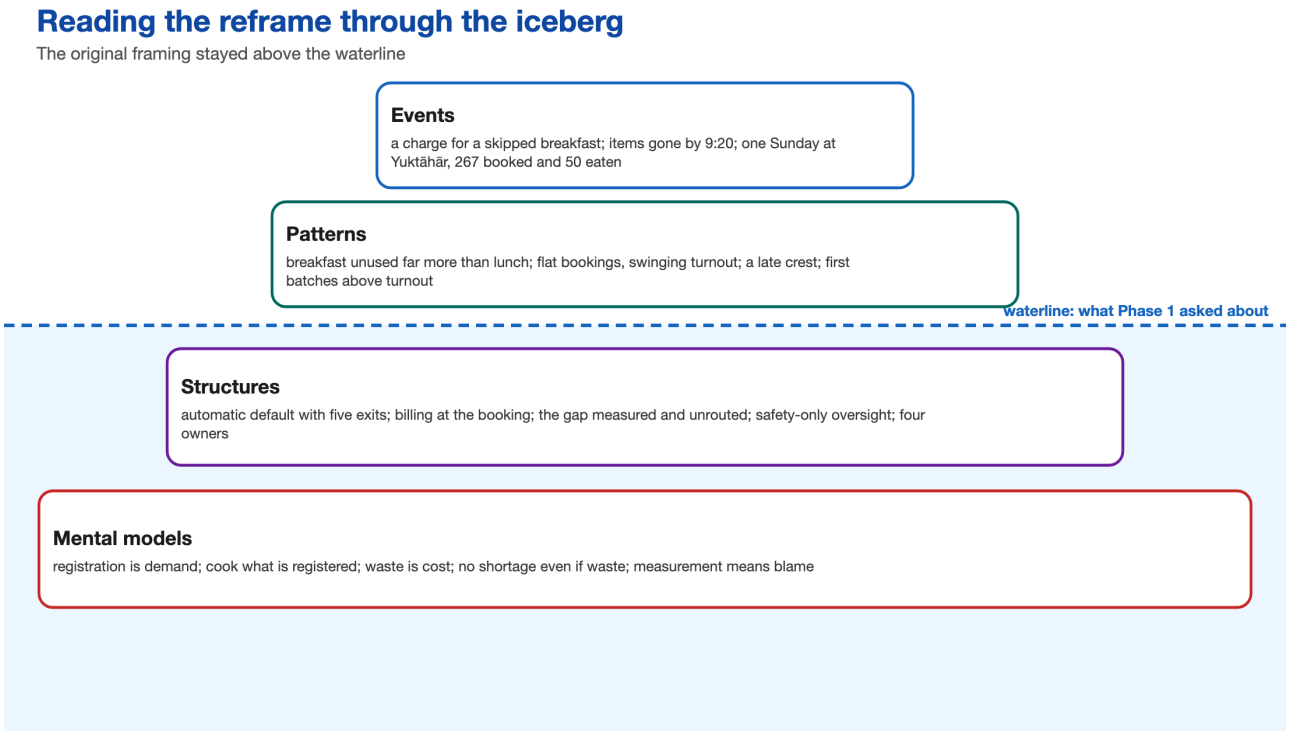



*Figure 5. The paradox read at four depths.*

**Events.** A student charged Rs. 48 for a Kadamba breakfast they were booked for and skipped. Idli, puri and bhatura gone by 9:20 with promised restocks that do not arrive. Sunday 6 September at Yuktāhār: 267 breakfasts booked, 50 eaten (confirmed). A Kadamba biryani day when 25 students arrived after close and the door was reopened (single-sourced). The wrapped display plate at Kadamba's entrance on the day of the visit (confirmed). Yuktāhār dropping prediction software, and its staff reporting in December that waste had fallen since weighing began in August (single-sourced). A frog in the chicken biryani in November 2024, the December 2024 tightening of cancellation rules over a student survey and two Parliament meetings (secondhand; candidate), the Felicity walk-in relaxation, and the vote that ended the fixed-semester menu.

**Patterns.** Breakfast bookings go unused at a far higher rate than lunch, bookings stay flat across the week while turnout swings, and arrivals crest in the last quarter hour and spill past close. The first batch sits above turnout at every kitchen. The same items waste at Kadamba while others sell out, and at Bakul demand follows the item. All six interviewed students are on a standing booking, none got value from Skip Meal, and four use Mess Cell. Complaints sit until an incident forces a response, with one loud exception.

**Structures.** Billing fires at the booking. Registration is automatic with five exits. Each kitchen converts bookings into a first batch by its own rule, and cooked quantity can rise but never fall. Bakul's food is cooked at Hafizpet, carried about 8 km in insulated boxes held at 60 to 72 °C, and topped up by phone across the same distance (single-sourced). Walk-in pricing runs 70% to 90% above the registered rate. Skip Meal pays the student nothing, and whether it lowers what Kadamba cooks is in conflict (section 8). The committee chooses the menu while the operator sees the waste. Complaints route down into the operator and nothing routes measurements up. Oversight inspects safety and taste, never quantity. Menu, kitchen, timetable and billing have four owners and none owns the outcome. Feedback is triaged by loudness, not cost.

**Mental models.** That a registration count is a good enough stand-in for demand, held by the rule setters. That the operator's job is to cook what is registered (Kadamba), or that waste is a cost to drive down (Yuktāhār), or that there should be no shortage even if food is wasted (Bakul). That prediction belongs close to the operator's own data, or that beyond a fixed ratio there is nothing to predict with (”we don't have that idea”, Kadamba). That measuring waste means assigning blame, the fear Yuktāhār's managers had to remove. That breakfast turnout is higher than it is. That no-shows are students away with a friend eating in their place. That a registration already made costs nothing extra to leave unused. That planning certainty always beats demand accuracy. That operational policy belongs to the Mess Office alone. That breakfast is a meal every student should be placed into by default.

Phase 1 asked about events and patterns. Everything that would move this system sits in the bottom two rows.

## 5. Why This System Holds Steady Instead of Breaking

A problem usually persists because nobody solved it, solving it costs too much, or somebody benefits. None of those is the story here. This gap holds steady because every consequence it creates is absorbed by something that works, before it reaches anyone who sets the rules.

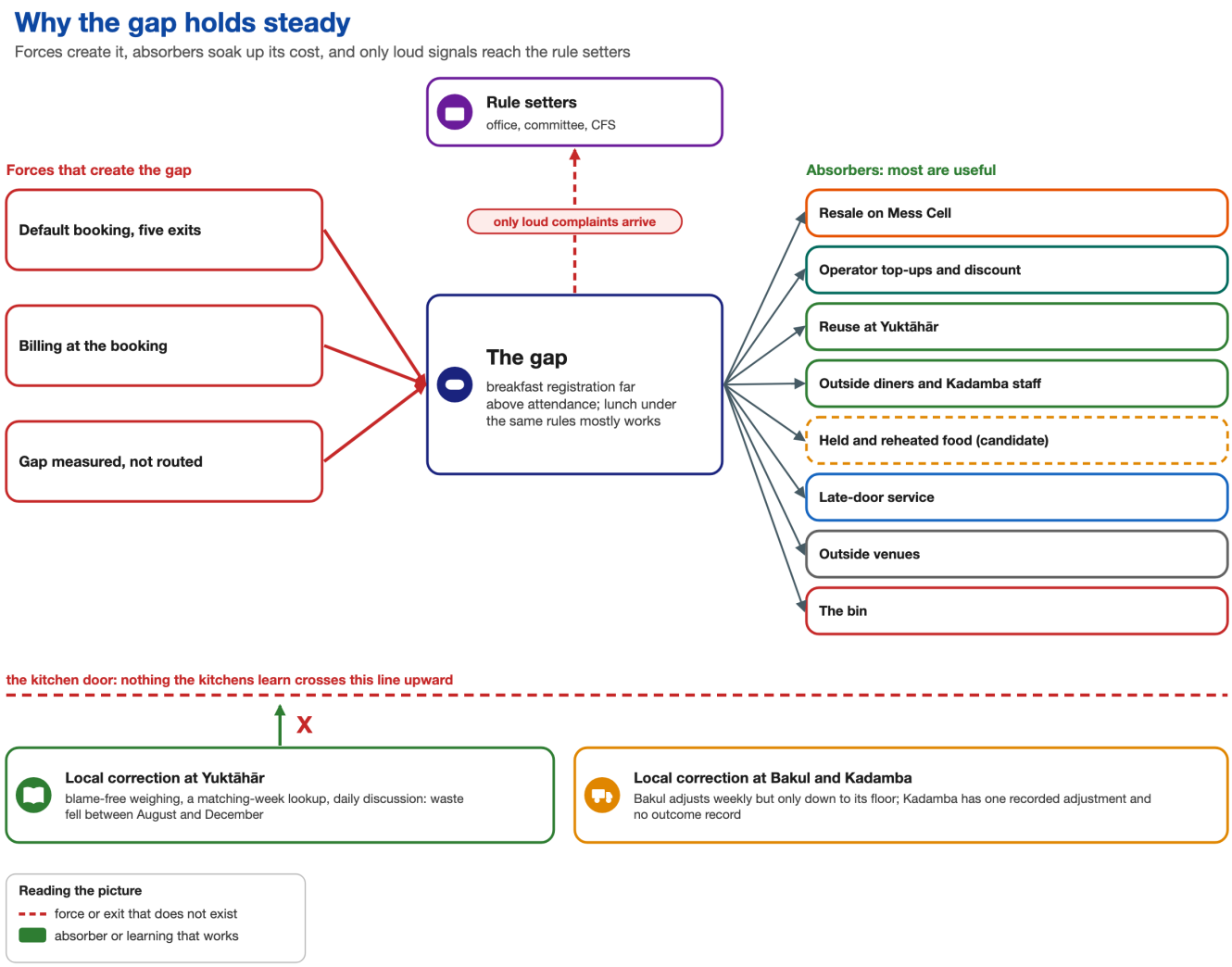



***Figure 6.*** *Why the gap holds steady: three forces create it, eight absorbers soak up its cost, and only a loud signal reaches the rule setters.*

Resale absorbs the student's financial loss. The largest absorber is the operator's own daily adjustment: each operator converts the booking into its own cooked quantity, watches the counter and tops up or re-cooks, which turns the gap into a kitchen routine nobody upstream must answer for. Yuktāhār reuses leftovers: idli becomes idli upma or idli poha for lunch, dosa batter goes back to the fridge, surplus milk is set as curd (confirmed). At Kadamba about 30 portions are planned for 30 to 50 outside diners such as night-shift security, and about half of Kadamba's staff eat from the same food (single-sourced), an uncounted sink for first-batch surplus (candidate). The operator has not said whether those portions sit inside the first batch or on top of it, and whether they are the same allocation as the 30 to 40 extra portions held for walk-ins is unconfirmed. The door reopened for latecomers absorbs part of the crest. Holding and reheating converts surplus into lower quality: in the team's judgement Kadamba's food is cooked ahead, reheated and moved between vessels until it degrades (candidate). Whether that drives skipping is not evidenced, so it is an open chain. At Bakul, food held for about 50 at 9:20 carries part of the first batch into the last minutes, and some leftovers go back to Hafizpet and the rest are thrown (single-sourced). Hot holding keeps temperature while aloo paratha softens in its own steam, and the reserve for late diners is the food held longest (candidate). Outside venues absorb frustration that would otherwise become a complaint about hours, and informal workarounds absorb the dietary and cultural mismatch that never gets filed as a request. What none of these can use leaves in the bin. Kadamba has no composting: its waste goes to a dustbin collected by a garbage contractor (single-sourced), which the institution's account of an outside compost facility has not yet been reconciled with.

Most of this helps people. Students recover money, outside diners and staff are fed, and surplus is used where it can be. Any change to the first batch must plan their portions. Three absorbers protect no one: the over-cooked first batch, repeated reheating and the bin. Reducing those is a legitimate target in a way that removing resale or staff meals is not.

Together the absorbers mean no financial, volume or cultural signal reaches anyone positioned to act. At least one kitchen does weigh plate waste and kitchen waste item by item and records when each item ran out (confirmed), but that measurement never enters a rule-setting forum, and the system above the operator listens for complaint intensity, not records.

That changes the question from why this has not been fixed to what would make someone fix it. Something already does, locally. Yuktāhār cut its waste between August and December by weighing every meal, discussing it daily and making the weighing blame-free after staff feared blame, under managers who treat waste as cost against a profit goal (single-sourced, tracking confirmed). Kadamba has one recorded adjustment, cutting ragi idli in favour of plain idli, and no outcome record. Bakul adjusts weekly but only down to its floor. The local correction that goes furthest depends on an operator whose culture points at waste and on measurement nobody fears, and where the culture points at shortage it stops at the floor. Nothing carries these fixes up to the rules that keep regenerating the gap.

## 6. Developing Alternative Problem Framings: Other Ways to Frame This, and Why They Fall Short

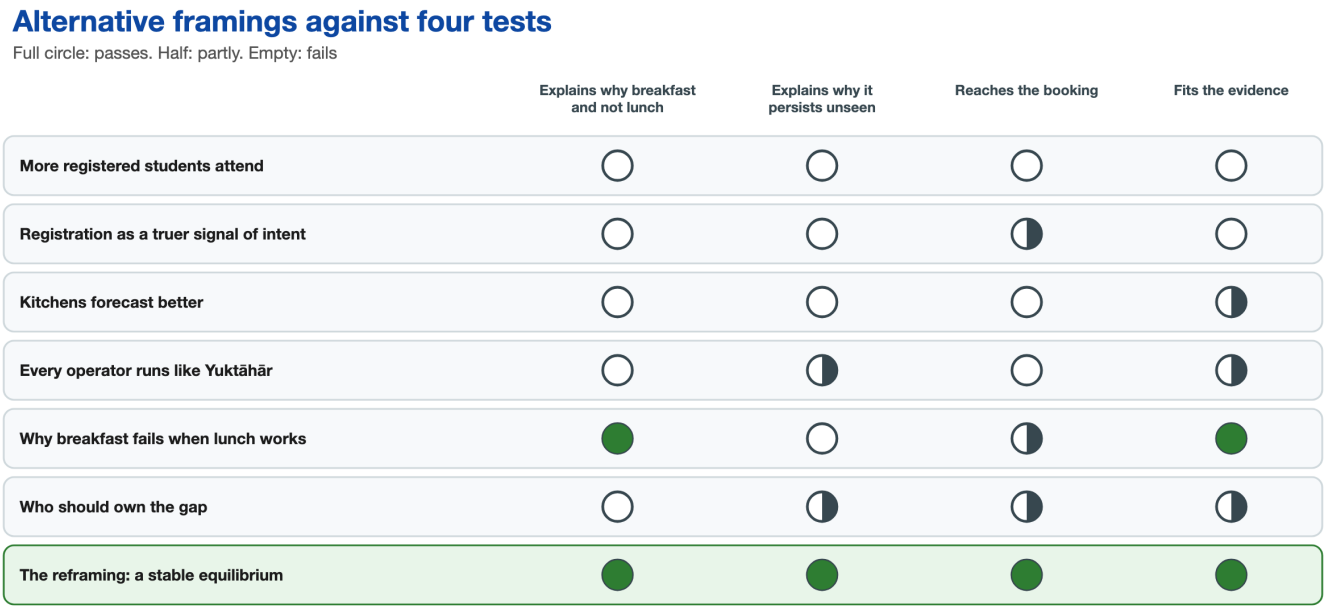



*Figure 7. Six alternative framings against four tests; only the reframing passes all four.*

**"How do we get more registered students to attend?"** This was the original framing. It treats attendance as the goal when a real share of non-attendance is legitimate, and it aims the fix at the people with the least power and the most defensible behaviour.

**"How do we make registration a more accurate signal of intent?"** Registration cannot express intent when it is assigned automatically, and the institution's own proven alternative needed no change in student behaviour. It removed the need for registration to be accurate.

**"How do we help kitchens forecast better?"** The operators already forecast and adapt, and the best-instrumented kitchen still believes up to 50% where its register says 28%, cannot un-cook a batch, and projects from a first half hour that misses the crest. Bakul already adjusts weekly and keeps food above its manager's own highest estimate

because its goal prefers waste to a run-out. A better forecast of a default booking cannot see a morning decision the booking never records.

**”How do we get every operator to run like Yuktāhār?”** Closer, because culture decides whether the gap is measured at all. It leaves automatic registration, billing at the booking and the missing upward route untouched, and Yuktāhār itself still overestimates breakfast. Culture can reduce the waste and not the gap.

**”Why does breakfast fail when lunch works under the same rules?”** Closer again. It isolates a default booking against a volatile morning decision, a late crest and legitimate skipping, but cannot say why so large and well recorded a difference never reaches a rule owner.

**”Who should own the gap?”** Fragmented ownership explains why a known fix was not picked up, and the operator owns the gap's consequences without owning any of its causes. It does not explain why the gap never got loud enough to demand an owner.

The framing at the top of this document is the one the evidence supports: a stable equilibrium specific to breakfast, held by a default booking, billing on that booking and a missing route from those who measure to those who set the rules, and protected by absorbers that are mostly useful.

## 7. Validating the Reframing with Evidence

The statement was tested against every assumption the project started with. Most were revised or overturned.

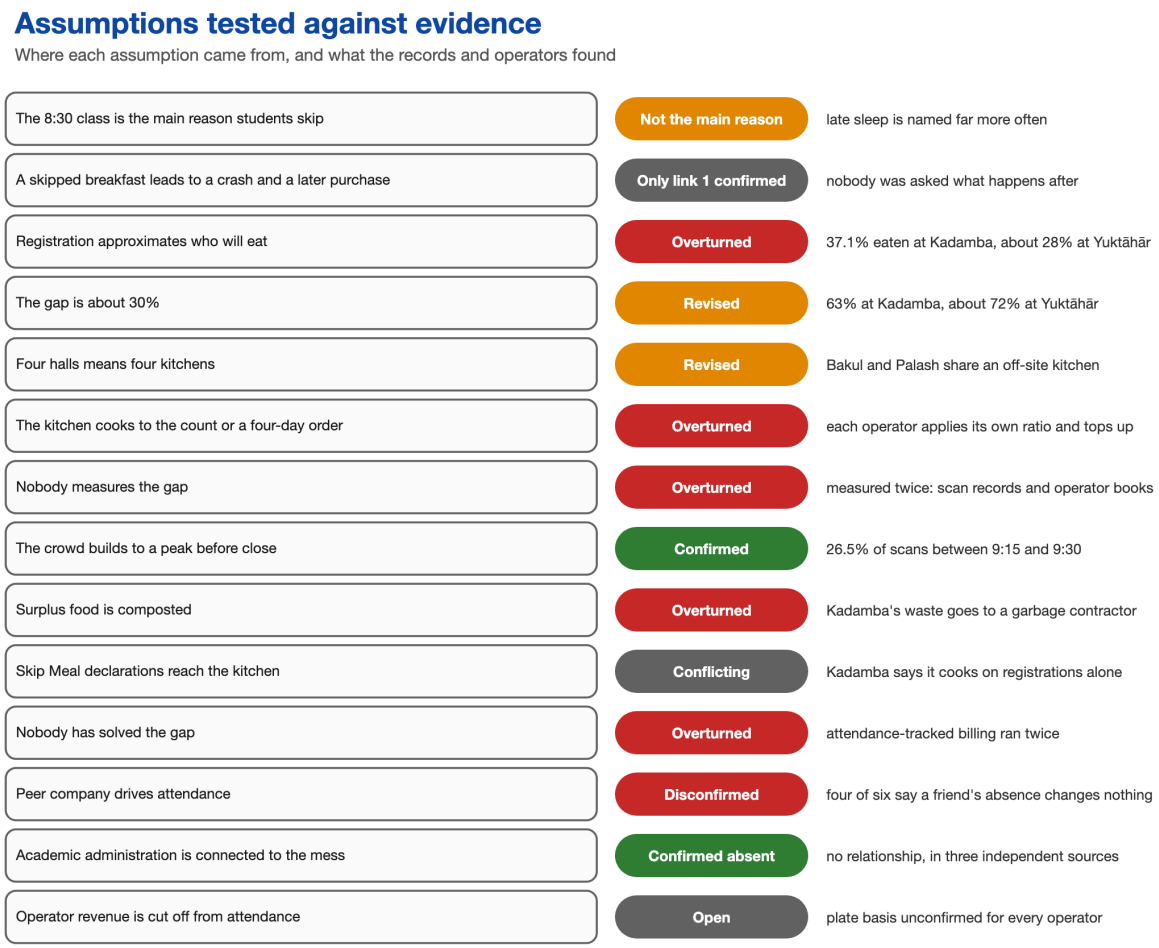



*Figure 8. Assumptions tested against evidence: what held, what changed, what is open.*

Seven pieces of evidence support the framing and one open question limits it.

- **A fix is possible.** The attendance-tracked model ran twice at this institution and closed the exact gap. The scan data it needs is captured daily under the standing model.
- **The booking is a default.** Confirmed three ways: the CDS poster's cancellation rules, a student's account of automatic registration with no way to nullify the charge, and Kadamba bookings flat across every weekday.
- **The gap is large and measured.** Two halls, from records, not recall (section 1).
- **This is a breakfast problem and billing alone does not explain it.** Yuktāhār's lunch runs at 56% to 89% under the same rules on the days breakfast runs at 18% to 40%.
- **The system adapts locally.** Three operators independently describe converting the booking to a lower cooked quantity and topping up, each under a different regime.
- **Fragmentation explains why the fix was not kept.** The CFS Chair has known the turnout figures for years without them entering a registration or billing conversation, and Yuktāhār's weekday and per-item records have reached no rule owner either. Its believed 50% against a recorded 28% shows nobody checks one number against another.
- **Attention follows loudness.** Menu fatigue travelled from Student Council to Committee and changed policy. The larger measured gap has never entered that process. A vendor that regards the count as what it is paid for has no stated reason to push the gap upward.

The pattern matches the clearest in the project. The system optimises what it can measure and bill, the registration count. Its own stated goal, printed on the CDS dining poster, is nutritious, hygienic, affordable meals with minimal food wastage through advance meal planning, and the CDS handbook names reduced wastage and better meal planning as the benefits of registering early. The easy metric and the stated goal have diverged, and the institution already knows it.

One question stays open and has narrowed without closing: what a plate counts for in each operator's monthly payment. Bakul's manager says plates served and Kadamba's remark points to booked plates (both single-sourced). Contracts may differ by operator, ”served” may be a normalised word for billed plates, or Kadamba's remark may describe duty and not the payment formula. No contract has been seen. The answer decides who bears the cost of a no-show, whether operator revenue depends on the gap, and whether a standing attendance-tracked model shifts risk onto the operator. If the terms rule it out the framing still holds, and the fix moves from adoption to renegotiation. If Bakul is paid on served plates it already bears the cost of what it over-cooks and accepts it (candidate).

## 8. What's Still Missing

Operations are now covered: three operators were interviewed, Yuktāhār walked through every book, a first-hand account fixed the cancellation rule and the Kadamba command chain, and April's scans measure the crowd curve. What remains open is stated and not merged.

- **Plate basis.** Whether operators are paid on booked or served plates, and the deductions, contract by contract. Nobody has asked directly. It separates a considered trade-off from an unexamined default.
- **Skip Meal.** Kadamba says it cooks on registrations only, while an earlier finding says declarations reach the kitchen. Skip Meal may adjust the count the operator sees while the operator treats any count as the obligation. One direct question settles it.
- **CDS head.** The operator names one person and earlier research another, and it is open whether these are two roles or one wrong name. Whether the operators' ”CDS team” is the Mess Office is also open, and the CDS head has never been interviewed.
- **Registration cadence.** Operators describe weekly blocks and students describe monthly, and the actual rule is unconfirmed.
- **Kadamba waste.** The base of the 5% to 10% figure, whether any outcome record exists beyond gram standards, whether outside-diner portions and the walk-in buffer are one allocation, and whether outside diners are registered anywhere (”some sort of registration done for outside”).
- **Holding and reheating.** The chain rests on the team's judgement and has not been observed at service level, and whether the roughly 37 scans a day at or after 9:30 are the back-door arrivals needs one check.
- **Skip Meal's low use.** Awareness, trust and portal friction would each point to a different fix, and this project cannot yet tell them apart.

- **Waste route.** Where the garbage contractor takes Kadamba's waste, whether any is composted, and Yuktāhār's disposal after reuse. Disposal has not been observed.
- **Bakul.** Whether Palash shares its kitchen, whether the weekly review is written down, how long a Hafizpet top-up takes, whether the 9:20 reserve rests on data or judgement, which menu decisions the manager can override, and why residents skip.
- **Records and timetable.** Lunch, dinner and other-hall scans have not been seen, and whether the 9:15 to 9:30 crest lines up with a first class ending needs one check against the timetable.